Project Documentation

Overview

Project Name	Regenerative AI for Sustainable Agriculture: A Smart Crop Monitoring Framework
Project Manager	Kirti Vardhan Singh
Project Dates	Start Date: Jun 4, 2025 End Date: Jul 5, 2025
Background	Traditional farming methods often lack real-time insights into crop health, leading to reduced yield and inefficient resource usage. This project was initiated to develop an AI-powered, scalable, and accessible framework combining plant disease detection and NDVI-based crop health monitoring to assist farmers—especially smallholders—in achieving precision agriculture.
Objectives	• Build a CNN-based model for real-time plant disease classification using the PlantVillage dataset. • Develop an NDVI calculation module using Sentinel-2 satellite images. • Integrate both features into a user-friendly Streamlit app. • Enable offline inference for edge deployment (e.g., Jetson Nano, Raspberry Pi). • Empower farmers with actionable insights to improve crop yield and sustainability.

Target Audience	• Smallholder and large-scale farmers seeking precision agriculture tools. • Agricultural researchers interested in AI-powered monitoring. • Government bodies or NGOs promoting smart farming. • Students and educators exploring AI applications in sustainability.

Project Specifics

Project Scope	The project includes training a CNN model, developing an NDVI heatmap generator, and creating a unified web app using Streamlit. It involves satellite imagery processing, deep learning, and UI development. The scope also includes experimentation with local/edge deployment and limited dataset packaging for GitHub.
Project Constraints	• Large datasets (PlantVillage & satellite images) could not be pushed to GitHub. • Hardware limitations for real-time edge deployment. • Limited access to real-time IoT field data; used simulated or public data. • Internet dependency for full Sentinel-2 access (can be mitigated by offline tiles).
Deliverables	• Trained CNN model for plant disease classification (plant_disease_cnn_model.h5). • NDVI GeoTIFF parser and visualizer module (ndvi_tif.py). • Fully working Streamlit web app (app.py). • CNN training notebook and preprocessing scripts. • README documentation and GitHub repository. • Research paper detailing methodology and results.
Explorations & Decisions	Initially considered using ResNet or MobileNet, but opted for a lightweight custom CNN for compatibility with edge devices. Explored both local and cloud NDVI rendering pipelines; chose Rasterio for Python-based local processing. Decided to use Sentinel-2 bands due to open availability and optimal resolution for NDVI.

Project Timeline

Task or Deliverable	*Owner*	*Date Completed*	*Notes*
CNN model training	Kirti Vardhan Singh	Jun 5, 2025	Trained using PlantVillage dataset
NDVI module creation	Kirti Vardhan Singh	Jun 10, 2025	Implemented using Rasterio and Numpy
Streamlit app integration	Kirti Vardhan Singh	Jun 14, 2025	Combined CNN and NDVI features
Final testing and deployment	Kirti Vardhan Singh	Aug 25, 2025	Demoed on local server
Research paper documentation	Kirti Vardhan Singh	Jul 4, 2025	Includes model details and evaluations

Conclusion

Project Outcomes	The project successfully demonstrated how AI and remote sensing can work together to empower agriculture. The model achieved high accuracy on validation sets, and the NDVI module accurately identified vegetation health zones. The app was easy to use, with potential for real-world deployment.

Recommendations	• Integrate IoT sensors for soil moisture and pH. • Expand dataset to cover local/regional crops. • Add support for voice commands and local language translations. • Deploy as a mobile app with low-latency NDVI preview using tiles. • Test real-time field deployment with hardware like Jetson Nano.
Resources	• GitHub: [Insert your GitHub link] • Research Paper: Available upon request • Contact: kirtivardhan7549@gmail.com • Dataset: PlantVillage:https://www.kaggle.com/datasets/emmarex/plantdisease Sentinel-2: https://apps.sentinel-hub.com/eo-browser